Def. Let R be an integral domain.
A Positive norm N on R is called a dedekind-hasse norm if: for any non-zero $a, b \in R$, either $a \in (b)$ or for some $d \in (a,b)$ satisfies $0 < N(d) < N(b)$

$\Leftrightarrow$ d is nonzero.

Note that: N is a Positive Euclidean Norm means that given non-zero $a, b \in R$,

$$a = qb + r \text{ for some } q, r \in R \text{ with } r = 0 \text{ or } N(r) < N(b).$$

If $r = 0$, then $a = qb$, that is $a \in (b)$.
If $r \neq 0$, then $a = qb + r$ with $0 < N(r) < N(b)$, that is $0 < N(a - qb) < N(b)$.
Clearly, $a - qb \in (a,b)$, so it is a particular case of the dedekind-hasse norm.

Prop. Let R be an integral domain.
R is a Principal ideal domain <u>if and only if</u> R has a Dedekind-Hasse norm.

Proof Suppose that R is a P.I.D. Then, automatically R is U.F.D. Define

$$N : R \to \mathbb{N} \cup \{0\} : a \mapsto \begin{cases} 0 & \text{if } a = 0 \\ 1 & \text{if } a \text{ is unit} \\ 2^n & \text{if } a \text{ has } n \text{ irreducible factors.} \end{cases}$$

Since R is U.F.D., this N is well-defined.┘
And, given $a, b \in R$, $N(ab) = N(a) \cdot N(b)$, because:
If $a = 0$, then $N(ab) = N(0) = 0 = 0 \cdot N(b) = N(a) \cdot N(b)$. Similarly, if $b = 0$, so is,
If a is unit, then $N(ab) = N(b) = 1 \cdot N(b) = N(a) \cdot N(b)$, since the factorization of b is unique up to multiplying of any unit. Similarly, b is unit, so is,
If both a and b are non-zero non-unit elements, Then the uniqueness of factorization gives the multiplicative. So N is multiplicative.┘
To show that this N is a dedekind-hasse norm, let $a, b \in R$ be non-zero elements.
Since R is a P.I.D., $(a,b) = (r)$ for some $r \in R$.
If $a \notin (b)$, then $r \notin (b)$ because $a \in (b)$ and $r \notin (b)$, then $(a,b) = (b) \neq (r)$, contradict.
So that r is not divided by b, that is, $b = xr$ for some non-unit nonzero $x \in R$.
Now, $N(b) = N(xr) = N(x)N(r) > N(r) > 0$, so N is multiplicative dedekind hasse norm.┘
Conversely, let N be a dedekind-hasse norm on R. Let $I \subseteq R$ be a non-zero ideal.
By the well-ordering Principle, Choose a non-zero $b \in I$ s.t $N(b)$ is minimal.
Then, given $a \in I$, $(a,b) \subseteq I$ and the minimality gives $a \in (b)$, so $I \subseteq (a,b) = (b)$. ☒

「Example.

Let $R = \mathbb{Z}\left[\frac{1+\sqrt{-19}}{2}\right]$ be the quadratic integer ring